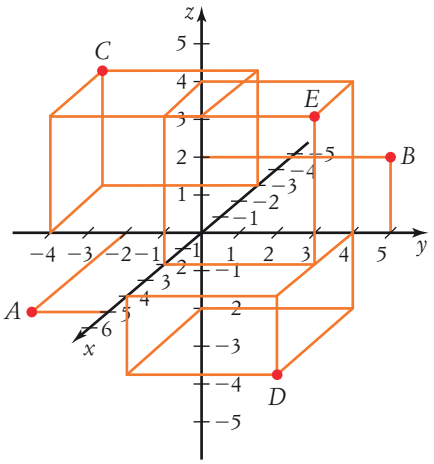
WORKSHEET

Chapter 1: Distances, midpoints and spheres

Use the following diagram to answer Questions 1 and 2.



1 Calculate the distance between each of the following pairs of points:

a A and D

b C and E

c E and B

d A and B

e C and D

© Cengage Learning Australia Pty Ltd 2014

f A and E

2 Calculate the midpoint of the following pairs of points:

a C and A

b B and C

c D and B

d A and D

e E and A

f A and B

3 For each of the following, calculate the distance AB.

a $A(0, 1, 2)$, $B(2, 3, 1)$

b $A(1, -2, 3)$, $B(-2, 2, 1)$

c $A(3, 4, 1)$, $B(2, -3, -5)$

© Cengage Learning Australia Pty Ltd 2014

d $A(4, 0, 7), B(0, 2, -6)$

e $A(3, 2, 1), B(1, -2, -3)$

f $A(-3, 2, -4), B(2, -3, 5)$

4 For each situation in Question 3, calculate the midpoint of AB.

a

b

c

d

e

f

5 In each of the following cases, find the equation of the sphere whose centre (C) and radius (r) is given.

a $C = (0, 0, 0), r = 2$

b $C = (0, 2, 0), r = 3$

c $C = (-1, 4, 0), r = \sqrt{5}$

d $C = (2, 1, -3), r = 5$

© Cengage Learning Australia Pty Ltd 2014

e $C = (1, -3, 2), r = 2\sqrt{3}$

f $C = (4, 2, -3), r = \frac{1}{2}$

6 In each of the following, find the centre and the radius of the sphere.

a $x^2 + y^2 + z^2 - 25 = 0$

b $x^2 + y^2 + z^2 - 4x + 6y - 3 = 0$

c $4x^2 + 4y^2 + 4z^2 + 8x - 4y - 3 = 0$

d $x^2 + y^2 + z^2 + 2x - 8z + 5 = 0$

e $x^2 + y^2 + z^2 - 6x + 4y - 10z - 7 = 0$

f $x^2 + y^2 + z^2 - 2x - 6y - 14z + 13 = 0$

7 Find the equation of a sphere with centre (0, 0, 0) that passes through the point (2, 3, 6).

8 Find the equation of a sphere with centre (2, 3, −1) that passes through the point (4, −1, 1).

© Cengage Learning Australia Pty Ltd 2014

9 A sphere has a diameter with endpoints $(3, 5, -3)$ and $(7, 3, 1)$. Find the equation of the sphere.

10 The vertices of two triangles are given below. Determine if each triangle is scalene, right-angled or isosceles.

a $(1, 2, 3)$, $(0, 2, 4)$ and $(0, 0, 2)$.

b $(1, -3, -2)$, $(5, -1, 2)$ and $(-1, 1, 2)$.

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** $\sqrt{41}$

b $90 = 3\sqrt{10}$

c 3

d $\sqrt{78}$

e $\sqrt{138}$

f $\sqrt{61}$

2 **a** $\left(1, -3, \frac{3}{2}\right)$

b $\left(-\frac{3}{2}, \frac{1}{2}, \frac{5}{2}\right)$

c $\left(2, \frac{9}{2}, 0\right)$

d $\left(\frac{9}{2}, 1, -1\right)$

e $\left(\frac{7}{2}, 1, 2\right)$

f $\left(\frac{5}{2}, \frac{3}{2}, 1\right)$

3 **a** 3

b $\sqrt{29}$

c $\sqrt{86}$

d $3\sqrt{21}$

e 6

f $\sqrt{131}$

4 **a** $\left(1, 2, \frac{3}{2}\right)$

b $\left(-\frac{1}{2}, 0, 2\right)$

c $\left(\frac{5}{2}, \frac{1}{2}, -2\right)$

d $\left(2, 1, \frac{1}{2}\right)$

e $(2, 0, -1)$

f $\left(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

5 **a** $x^2 + y^2 + z^2 - 4 = 0$

b $x^2 + y^2 + z^2 - 4y - 5 = 0$

c $x^2 + y^2 + z^2 + 2x - 8y + 11 = 0$

d $x^2 + y^2 + z^2 - 4x - 2y + 6z - 11 = 0$

e $x^2 + y^2 + z^2 - 2x + 6y - 4z + 2 = 0$

f $4x^2 + 4y^2 + 4z^2 - 32x - 16y + 24z + 115 = 0$

6 **a** $C = (0, 0, 0), r = 5$

b $C = (2, -3, 0), r = 2$

c $C = (-2, 0, 0), r = 3$

d $C = (-1, 0, 4), r = 2\sqrt{3}$

e $C = (3, -2, 5), r = 3\sqrt{5}$

f $C = (1, 3, 7), r = 6$

7 $x^2 + y^2 + z^2 - 49 = 0$

8 $x^2 + y^2 + z^2 - 4x + 2y - 2z - 18 = 0$

9 $x^2 + y^2 + z^2 - 10x - 8y + 2z + 33 = 0$

10 **a** Right-angled triangle

b Isosceles triangle

© Cengage Learning Australia Pty Ltd 2014